Regional Quantitative Problem-Solving Circle

Solution Key 2: Discrete Probability & Linearity of Expectation (Weeks 3–4)

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Rigorous Proofs & Structural Solutions

Problem 1 (Indicator Decomposition / Circle Symmetry)**.** N individuals sit in a circle; each independently points left or right with probability 1/2. Derive the expected number of unpointed individuals using indicator random variables.

Proof. Let X be the random variable representing the total number of unpointed individuals. We decompose X into a sum of N local indicator variables. For each individual $k \in \{1, \ldots, N\}$, define I_k such that:

$$I_k = \begin{cases} 1 & \text{if individual } k \text{ is unpointed} \\ 0 & \text{otherwise} \end{cases}$$

Thus, $X = \sum_{k=1}^{N} I_k$. To evaluate $\mathbb{E}[I_k]$, we calculate the marginal probability that individual k is not pointed at. The only people capable of pointing at k are their immediate left neighbor and immediate right neighbor. For k to be safe, the left neighbor must point left (probability 1/2), and the right neighbor must point right (probability 1/2). Since these two pointing actions are fully independent coin flips:

$$\mathbb{E}[I_k] = \mathbb{P}(I_k = 1) = \left(\frac{1}{2}\right) \times \left(\frac{1}{2}\right) = \frac{1}{4}$$

By the Linearity of Expectation, the expectation of the sum is the sum of the expectations:

$$\mathbb{E}[X] = \mathbb{E}\left[\sum_{k=1}^{N} I_k\right] = \sum_{k=1}^{N} \mathbb{E}[I_k] = \sum_{k=1}^{N} \frac{1}{4} = \frac{N}{4}$$

Note on Independence: The events $I_k = 1$ and $I_{k+1} = 1$ are highly dependent (if k is unpointed, it influences the direction the right neighbor pointed, altering the safety of $k+1$). However, the Linearity of Expectation operator ($\mathbb{E}[A + B] = \mathbb{E}[A] + \mathbb{E}[B]$) is an absolute axiomatic property of integrals and finite sums that holds structurally regardless of any joint covariance or dependence between the variables. □

Problem 2 (Permutation Fixed Points & Dependent Cycle Statistics)**.** A permutation $\pi \in S_n$ is chosen uniformly at random. Let X be the number of fixed points ($\pi(k) = k$) and Y be the number of transpositions (cycles of length 2). Compute $\mathbb{E}[X]$ and $\mathbb{E}[Y]$.

Proof. **Part 1 (Expected Fixed Points):** Let $I_k = 1$ if $\pi(k) = k$, and 0 otherwise. Thus $X = \sum_{k=1}^{n} I_k$. To compute $\mathbb{E}[I_k]$, we find the probability that element k is fixed. If $\pi(k) = k$, the remaining $n - 1$ elements can be arranged in $(n-1)!$ ways. Since there are $n!$ total permutations:

$$\mathbb{E}[I_k] = \frac{(n-1)!}{n!} = \frac{1}{n}$$

By Linearity of Expectation: $\mathbb{E}[X] = \sum_{k=1}^{n} \mathbb{E}[I_k] = n \times \frac{1}{n} = 1$.

Part 2 (Expected Transpositions): A transposition is an unordered pair $\{i, j\}$ such that $\pi(i) = j$ and $\pi(j) = i$. There are exactly $\binom{n}{2}$ such pairs. Let $J_{\{i,j\}} = 1$ if $\{i, j\}$ forms a 2-cycle, and 0 otherwise. Thus $Y = \sum_{\{i,j\}} J_{\{i,j\}}$. If $\pi(i) = j$ and $\pi(j) = i$, the remaining $n - 2$ elements can be freely permuted in $(n - 2)!$ ways. The expectation of any single pair indicator is:

$$\mathbb{E}[J_{\{i,j\}}] = \frac{(n-2)!}{n!} = \frac{1}{n(n-1)}$$

By Linearity of Expectation, we sum this over all $\binom{n}{2}$ possible pairs:

$$\mathbb{E}[Y] = \binom{n}{2} \times \frac{1}{n(n-1)} = \frac{n(n-1)}{2} \times \frac{1}{n(n-1)} = \frac{1}{2}$$

□

Problem 3 (Overcounting & Product Space Inclusions)**.** Independent integer selection from $\{1, 2, \ldots, N\}$ analyzing the probability of index-power relations where neither integer is a square of the other.

Proof. We select a pair (x, y) uniformly at random from the product space $\{1, 2, \ldots, N\} \times \{1, 2, \ldots, N\}$. The total number of possible pairs is N^2. Let A be the event that $x = y^2$, and B be the event that $y = x^2$. We want to find the complementary probability: $1 - \mathbb{P}(A \cup B)$.

To find the cardinality $|A \cup B|$, we apply the Principle of Inclusion-Exclusion:

$$|A \cup B| = |A| + |B| - |A \cap B|$$

The event A ($x = y^2$) occurs for every integer y such that $y^2 \leq N$. Since $y \geq 1$, there are exactly $\lfloor \sqrt{N} \rfloor$ such choices for y, which deterministically fixes x. Thus, $|A| = \lfloor \sqrt{N} \rfloor$. By symmetry, $|B| = \lfloor \sqrt{N} \rfloor$.

The intersection event $A \cap B$ requires both $x = y^2$ and $y = x^2$. Substituting the first into the second yields $y = (y^2)^2 = y^4$. Because $y \in \{1, 2, \ldots, N\}$, the only integer solution to $y = y^4$ is $y = 1$. This implies $x = 1^2 = 1$. Thus, the intersection contains exactly one unique pair: $(1, 1)$. Therefore, $|A \cap B| = 1$.

Substituting these counts back into our Inclusion-Exclusion formula:

$$|A \cup B| = \lfloor \sqrt{N} \rfloor + \lfloor \sqrt{N} \rfloor - 1 = 2\lfloor \sqrt{N} \rfloor - 1$$

The number of valid pairs where neither is the square of the other is the complement: $N^2 - (2\lfloor \sqrt{N} \rfloor - 1)$. Dividing by the total size of the sample space N^2, the exact probability is:

$$\mathbb{P} = \frac{N^2 - 2\lfloor \sqrt{N} \rfloor + 1}{N^2}$$

□